$Var(ax+by) = a^2 Var(x) + b^2 Var(y) + 2ab Cov(X,Y)$

Farm ← Cluster of barns, lvl. 2

Barn ← Vat, level 1

mix of feed & antibiotic ← factors leading to dependent response

HW4 1) Diet: Fixed Poultry Farm: Random

Antibiotics: Fixed Barn: ~~Subject~~

2) Storage temp: Fixed Dependent: Bacterial counts

Storage units: Random Crossed: Seafood & temp

Seafood type: Fixed Random: Storage unit

3) a) Repeated measure data, measurement = Lvl 1, rat = lvl 2

b) rat ID = random, region = fixed

c) see R PDF

4) a) $Math_{ij} = \gamma_{10}(SES_{ij} - \overline{SES_j}) + \gamma_{20}(female_{ij}) + \gamma_{30}(minority_{ij}) + R_{ij} + \gamma_{00}$ } fixed

$+ \gamma_{01}\overline{SES_j} + V_{0j}$ } Random

b) $Math_{ij} = \gamma_{10}(SES_{ij} - \overline{SES_j}) + \gamma_{20}(female_{ij}) + \gamma_{30}(minority_{ij}) + \gamma_{00}$ → Mean

Var: $\sigma^2 + \tau_0^2$ sd: $\sqrt{\sigma^2 + \tau_0^2}$

5) a) LVL 1: $science_{ij} = \beta_{0j} + \beta_{0j}(math_{ij}) + R_{ij}$ → $R_{ij} \sim N(0, \sigma^2)$

LVL 2: $\beta_{0j} = \gamma_{00} + V_{0j}$; $\beta_{1j} = \gamma_{10}$ → $V_{0j} \sim N(0, \tau_0^2)$

b) $science_{ij} = \gamma_{00} + \gamma_{10}(math_{ij}) + R_{ij} + V_{0j}$

c) $science_{ij} = \gamma_{10}(math_{ij}) + \gamma_{00}$ → mean

$Var = \sigma^2 + \tau_0^2$ $sd = \sqrt{\sigma^2 + \tau_0^2}$

6) a) LVL 1: $science_{ij} = \beta_{0j} + \beta_{1j}(math_{ij} - \overline{math_{ij}}) + R_{ij} \rightarrow R_{ij} \sim N(0, \sigma^2)$

LVL2: $\beta_{0j} = \gamma_{00} + \gamma_{01}(\overline{math_j}) + U_{ij}$; $\beta_{1j} = \gamma_{10} \rightarrow U_{ij} \sim N(0, \tau_0^2)$

b) $science_{ij} = \gamma_{00} + \gamma_{10}(math_{ij} - \overline{math_{ij}}) + R_{ij}$ } fixed

$+ \gamma_{01}(\overline{math_j}) + U_{ij}$ } random

c) $science_{ij} = \gamma_{10}(math_{ij} - \overline{math_{ij}}) + \gamma_{01}(\overline{math_j}) + \gamma_{00} \rightarrow$ Mean

$Var = \sigma^2 + \tau_0^2$ $sd = \sqrt{\sigma^2 + \tau_0^2}$